CIFAR-10 ACCURACY

```
history = model.fit(
    X_train,
    y_train_cat,
    validation_split=0.2,
    epochs=20,
    batch_size=64
)

loss, accuracy = model.evaluate(X_test, y_test_cat)

print("Test Accuracy:", accuracy)

plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.plot(history.history['accuracy'], label='Training')
plt.plot(history.history['val_accuracy'], label='Validation')
plt.title("Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.legend()

plt.subplot(1, 2, 2)
plt.plot(history.history['loss'], label='Training')
plt.plot(history.history['val_loss'], label='Validation')
plt.title("Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend()

plt.show()

predictions = model.predict(X_test)
predicted_classes = np.argmax(predictions, axis=1)
true_classes = y_test.flatten()

print(classification_report(true_classes, predicted_classes,
target_names=class_names))

cm = confusion_matrix(true_classes, predicted_classes)

plt.figure(figsize=(10, 8))
sns.heatmap(cm,
            annot=True,
            fmt='d',
            cmap='Blues',
            xticklabels=class_names,
            yticklabels=class_names)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```